

point with a fairly good community of people who had been trying it out. So, in all probability, if we have a problem or a question, somebody else in the community may have faced it earlier and could suggest a solution. It is a solid base to start from. It allows us to focus on the things that we do uniquely and gives us access to a community of people that we can rely on to help us if we need."

"Security was never a concern"

Many people worry about adopting open source technology or solutions because they believe security is a major concern. But for Belur, the security of open source technology came as the biggest advantage. He says, "Security is a huge concern for everyone. It is probably implicit to the choice of open source. Open source technologies are scrutinised by the best and the brightest in the world; therefore, a lot of security issues get addressed. We almost take that for granted, confident that our choice of open source allows us to say that we have an extremely solid and secure foundation. Our solutions are deployed by people on hardware in a network that is facing the Internet. So, they are subject to attack from a lot of different people."

Belur adds, "For example, a mail server may be deployed inside my lab, so it won't be attacked by a lot of people. In the end, it is within the boundaries of my firewall. Whereas, many of the products that we develop are deployed in a way that they are accessible on the Internet. So security is extremely important in such cases. I think open source technologies again come with an extremely high assurance of security. It doesn't mean there are no security problems, but they are found and fixed very rapidly. Users of open source get that advantage for sure."

Belur opines that it is courtesy the 'openness' of Linux-based technologies that security is not a prime concern. He points out, "The very model of open source is helpful in resolving the security issues. With the source code being accessible to all, people can thoroughly validate if the solution is secure because you can push your system to its limits and see what vulnerabilities are there to fix them, as opposed to buying Web server technology from Microsoft or any other proprietary software company, where you don't have access to the source code. In such cases, it is hard to identify the vulnerabilities and one has to wait for the vendors to acknowledge it, fix it and deliver a patch -- depending on how high-profile the issue is. The model of open source, on the other hand, lends itself to high security."

The team of engineers not only develops the company's solutions on open source technology but also uses Ubuntu Linux as the primary workstation. The reason is simple: ease of development. Belur explains, "We install Ubuntu Linux on our PCs simply because we believe that since our products are deployed on that platform, it would be easy for us to develop on the same platform."

Apart from that, open source technology is used to test the solutions as well. Some of the video-conferencing solutions that require the participation of hundreds of people can be tested with just a few mouse-clicks, thanks to open source technologies. Belur explains, "We also do a lot of internal testing. For instance, if we claim that over 1000 employees can watch a video-conference in

Some tips and tricks

Resorting to open source technologies to get the best solution and cut costs seems a great idea, but one has to be really cautious and choosy when it comes to settling for an open source technology solution. Belur shares some tips from his experience:

- I think one has to be very careful while choosing the technology. There are many open source technology projects. Some of them have a good community behind them and a good development roadmap ahead of them. It means that these projects will continue to get enhanced and evolve over a period of time. So if you pick such software as the building blocks of your solution, you can get huge productivity gains. But sometimes, you may end up choosing a wrong open source component—a dead-end project with not many people working on it. In projects that do not have much of a future, you tend to get stuck in the later stages. Such projects do not have an on-going development roadmap because they do not have a community around it. Such projects should be avoided. You just need to pick the winners.
- There can be times when you can get stuck in a strange jam with a problem that no one has ever encountered. Sometimes you have to spend hours solving the problem yourself because you are using the open source component in a way that no one else ever has. It means that you are open to the risk of having to go to the source of the project and try to solve a particularly strange problem. One should be mentally prepared for such issues.
- Last of all, you might have customers that say, "Can you produce me a security vulnerability report by a third party agency?" Fortunately, our customers are not like that, but if you are ever in such a situation, there are security agencies that do vulnerability and security analysis of products. They issue certificates for the tested products as well. Most companies that make solutions based on open source technology do not have such certifications available. Having certifications helps firms sell their solutions, particularly to enterprises. If there are issues related in getting the certification, then you will have to figure out on who will fix them.

which a company's chief makes an important announcement, we need to validate those claims. Yet, it's impossible to have 1000 people involved while testing such claims. So we use a lot of open source technologies that allow us to create the required number of virtual users. This makes the server feel that there are 1000 users logged in but that's not the case in reality."

"We use a lot of open source technologies in our QA processes as well. There are many common problems that software companies encounter. Open source technology is one platform on which people create solutions for these problems and they can be accessed by one and all. Sometimes, we use them to create our products, and at other times we use them as productivity tools to test our products," he adds. 

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